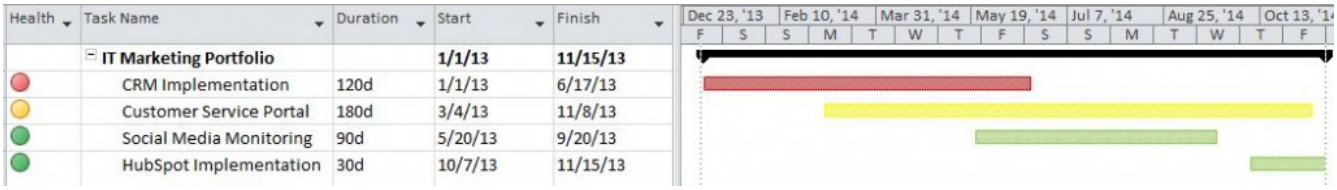
PART 1: Microsoft Project Dashboard

In this tutorial, you'll learn how to create the dashboard with red, yellow and green graphical indicators that automatically format the Gantt chart bars.



Health	Task Name	Duration	Start	Finish
	IT Marketing Portfolio		1/1/13	11/15/13
	CRM Implementation	120d	1/1/13	6/17/13
	Customer Service Portal	180d	3/4/13	11/8/13
	Social Media Monitoring	90d	5/20/13	9/20/13
	HubSpot Implementation	30d	10/7/13	11/15/13

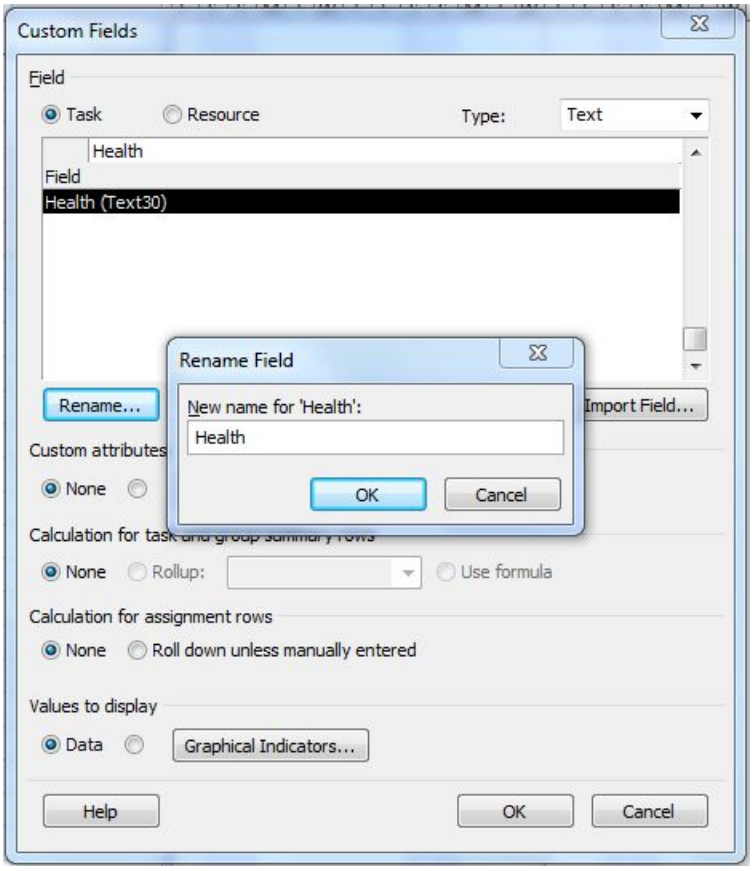
Step 1: Add a custom Health indicator

The first step is to customize an available Text field to display the "traffic light" graphical indicator. You need to modify an existing Text field with three valid values: Red, Yellow, and Green. In this example, I use Text30 however, you can use any unused Text field in Microsoft Project.

1. Insert the Text30 field.

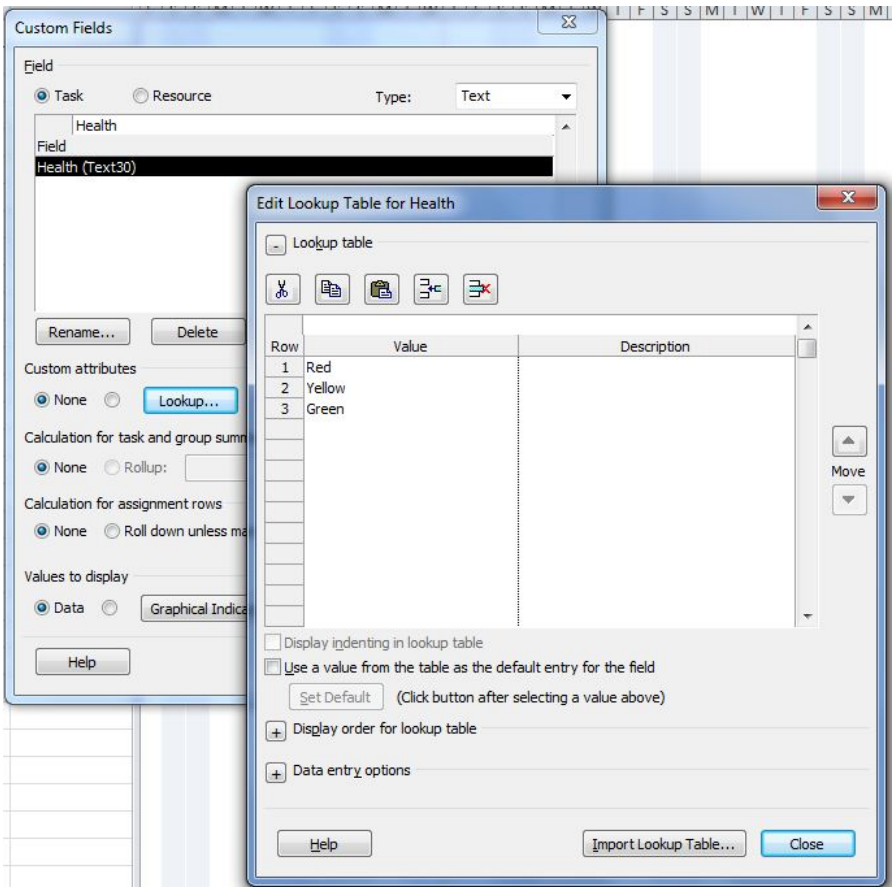
2. Right-click the Text30 field and select Customize Fields.

3. Click the Rename button and name the field Health.



4. Click the Lookup button.

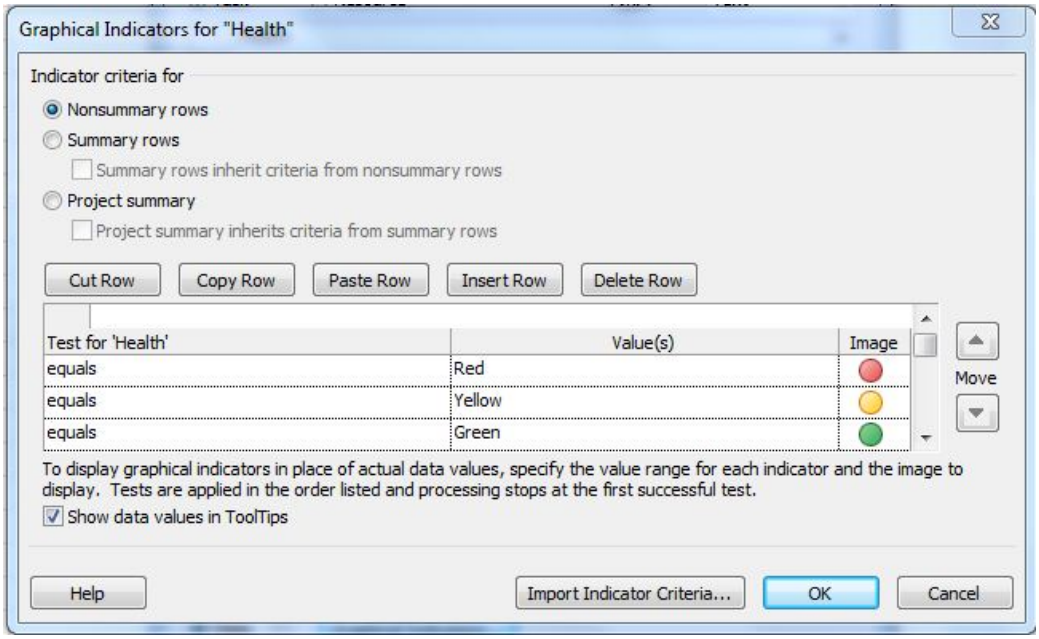
5. Add the values Red, Yellow, and Green.



6. Click Close.

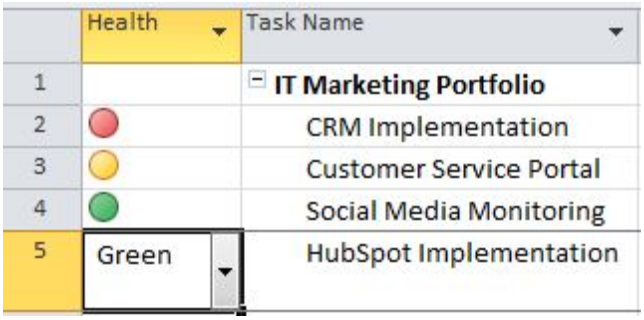
7. Click the Graphical Indicators button.

8. Add three tests for the Health field by entering “equals” and Red, Yellow, and Green for the appropriate value. Add a red, yellow, and green Image indicator.



9. Click OK.

By inserting the Health field into the Gantt Chart view, you can select from the Red, Yellow, or Green drop-down values, and the appropriate traffic light will be displayed.



The next step is to modify custom flag fields based on the Health indicator.

Step 2: Create the Late, At Risk, and On Target fields

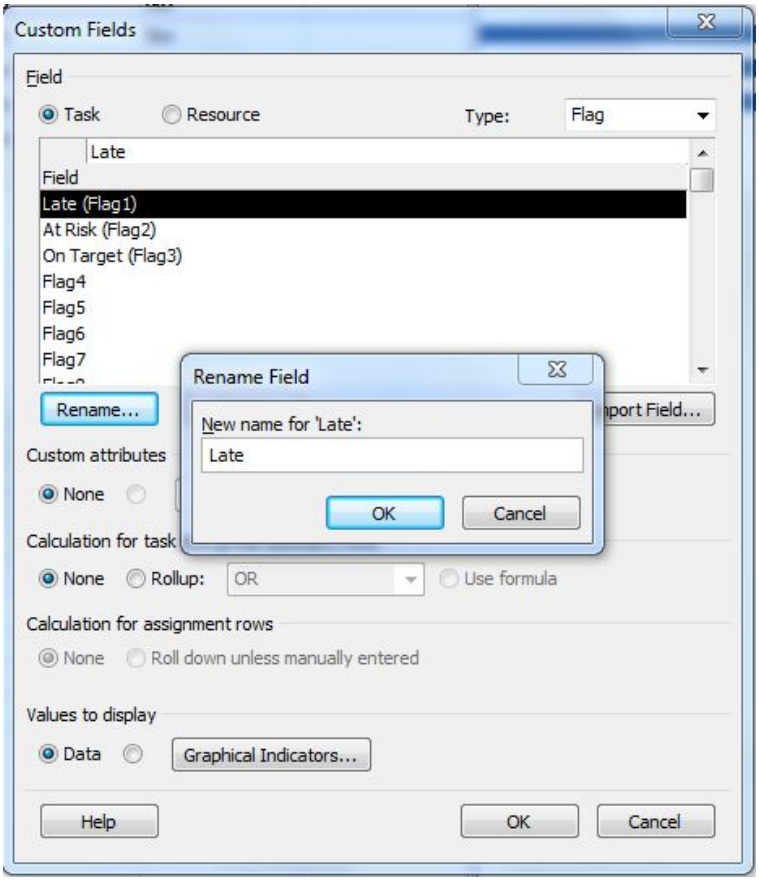
1. Insert three custom flags in the Gantt Chart view. I selected Flag1, Flag2, and Flag3.

2. Right-click the Flag1 field and select Customize Field.

3. Rename the Flag1 field to “Late.”

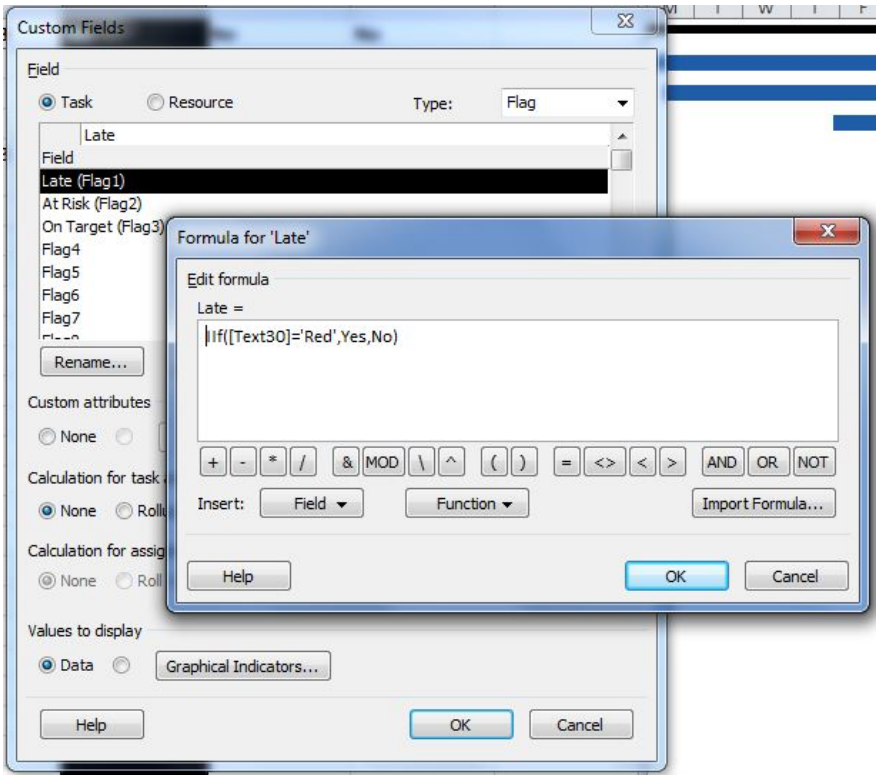
4. Rename the Flag2 field to “At Risk.”

5. Rename the Flag3 field to “On Target.”



6. Select the Flag1 field, click the Formula button, and enter the formula:

IIf([Text30]='Red',Yes,No)



7. Click OK.

8. Click the Flag2 field, click the Formula button, and enter the formula:

IIf([Text30]='Yellow',Yes,No)

9. Click the Flag3 field, click the Formula button, and enter the formula:

IIf([Text30]='Green',Yes,No)

10. Click OK.

By changing the Health graphical indicator field, the Flag1, Flag2, or Flag3 columns will change from No to Yes as appropriate.

	Health	Task Name	Duration	Start	Finish	Late	At Risk	On Target
		IT Marketing Portfolio		**1/1/13**	**11/15/13**	**No**	**No**	**No**
		CRM Implementation	120d	1/1/13	6/17/13	Yes	No	No
		Customer Service Port	180d	3/4/13	11/8/13	No	Yes	No
		Social Media Monitori	90d	5/20/13	9/20/13	No	No	Yes
		HubSpot Implementation	30d	10/7/13	11/15/13	No	No	Yes

11. Once you see the Late, At Risk and On Target columns change when the Health indicator changes, go ahead and hide the Late, At Risk and On Target columns.

Step 3: Create Red, Yellow, and Green bar styles

The next step is to create the specific Red, Yellow, and Green Gantt chart bar styles that will align to the graphical indicator color.
You can modify the Gantt chart bar styles to suite your preferences. In this example, we will add three new bar styles to match the Health indicator and the three flags.

1. Select Format | Bar Styles.

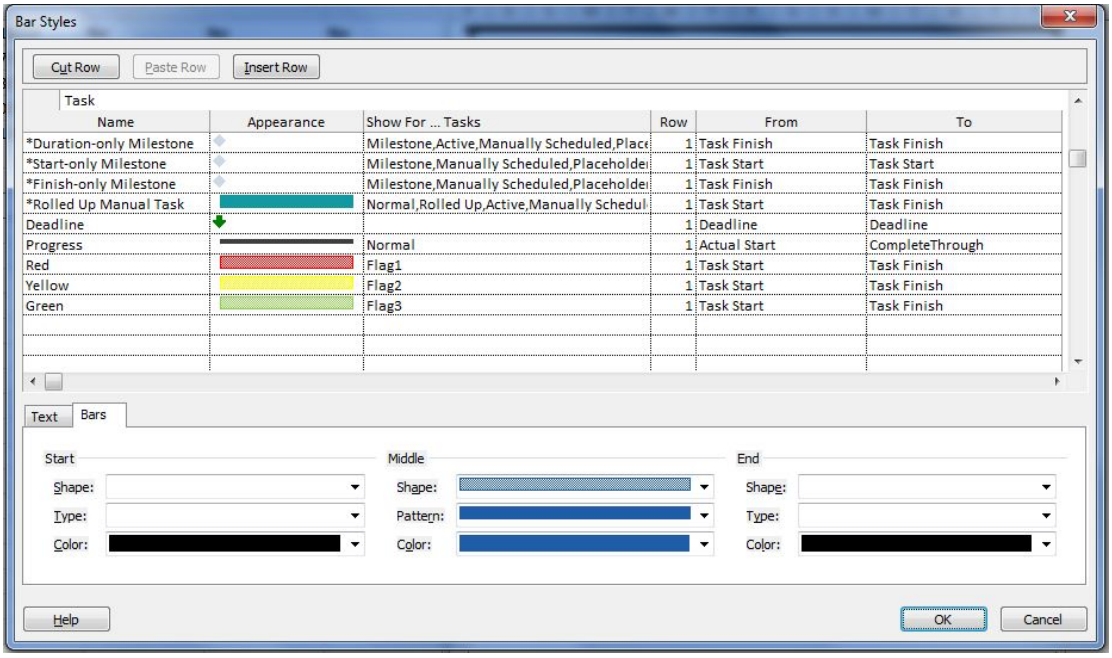
2. Insert three rows and name the rows Red, Yellow, and Green at the **bottom of the Bar Styles box**

The Bar Styles are processed in a hierarchical order so you need to move these styles to the bottom. Otherwise, another style can overwrite your custom style if the Show For … Tasks criteria matches.

3. On the Red row, click the Color drop-down menu and change it to Red.

4. Change the Show For … Tasks field to the Flag1 field. Repeat the step for Yellow and Green and assign Flag2 and Flag3 respectively. Remember to assign the yellow and green colors as appropriate.

5. Click OK



By following these first three steps, you've created a set of customized fields that will change the color of the Gantt Chart bar, depending on the Red, Yellow, or Green status.

Step 4: Enter project data

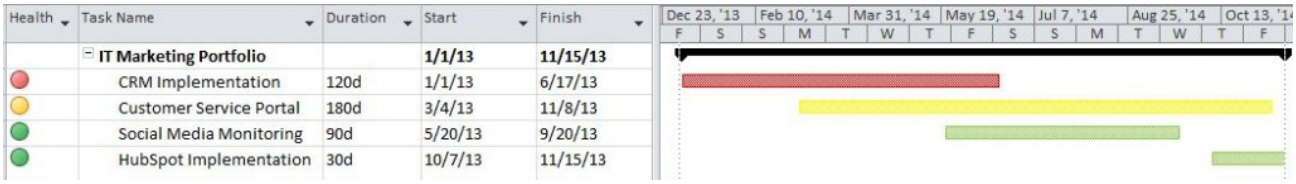
The next step is to enter the projects or work streams into the dashboard.

In this example, every project in your program or portfolio will be represented as a single task in the portfolio. Please note the the individual tasks existing in a different MS Project plan. You could role up all the project schedules to provide more detailed reporting. However, I've found for displaying program or portfolio health with timelines, creating a separate MS Project file is easier.

For a typical program or set of projects in the portfolio, I recommend entering the following data:

- Project Name
- Project Start Date
- Project Duration (which calculates the project finish)
- Complete
- Any dependencies between the projects as necessary

Step 5: Change the Health indicator based on status



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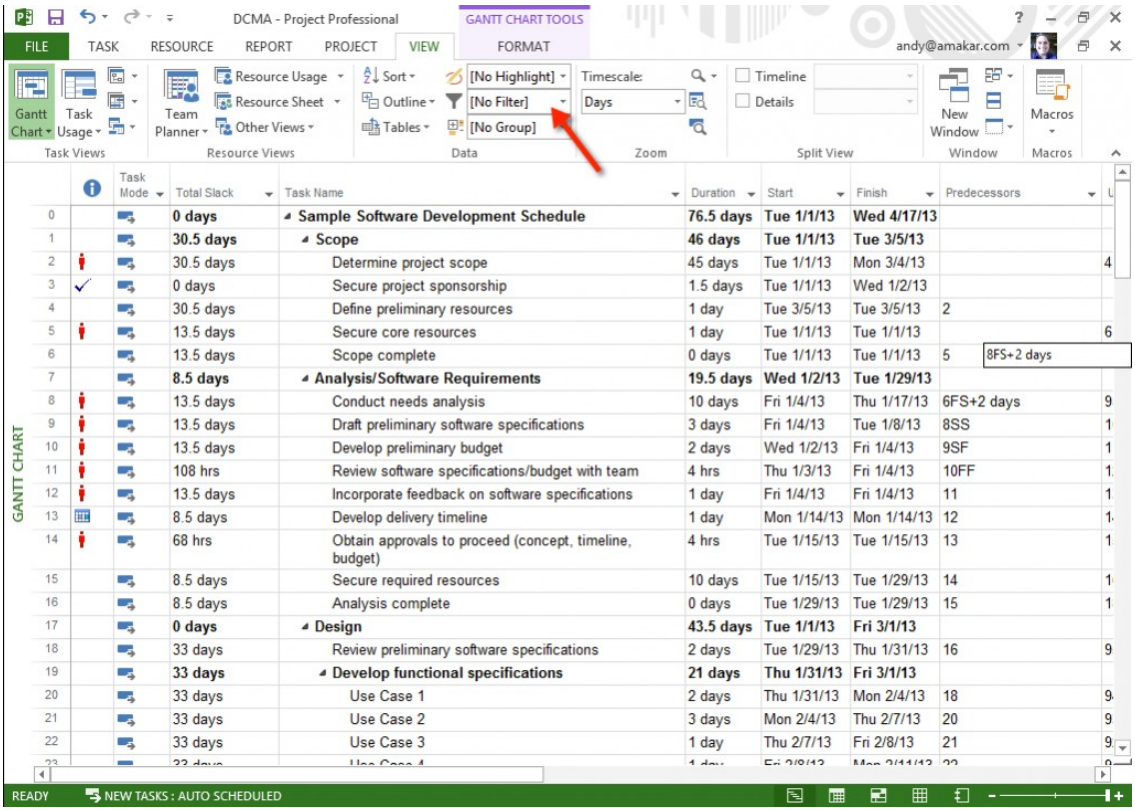
As you prepare for your next portfolio review or program status meeting, you can use this program dashboard to quickly communicate each project's status in the portfolio. You can also add various Microsoft Project data fields to the Gantt Chart table or customize the Gantt Chart further. To change the color, select the appropriate status from the Health field and the Gantt chart bar will change color.

Instead of drawing a separate Gantt chart or PowerPoint chart, project managers can use the Microsoft Project data to quickly represent status at a summary level. The dashboard can also be extended to assess new portfolio opportunities or identify specific projects that are funded, not started, or cancelled.

PART 2: Create Custom Filters with MS Project 2013

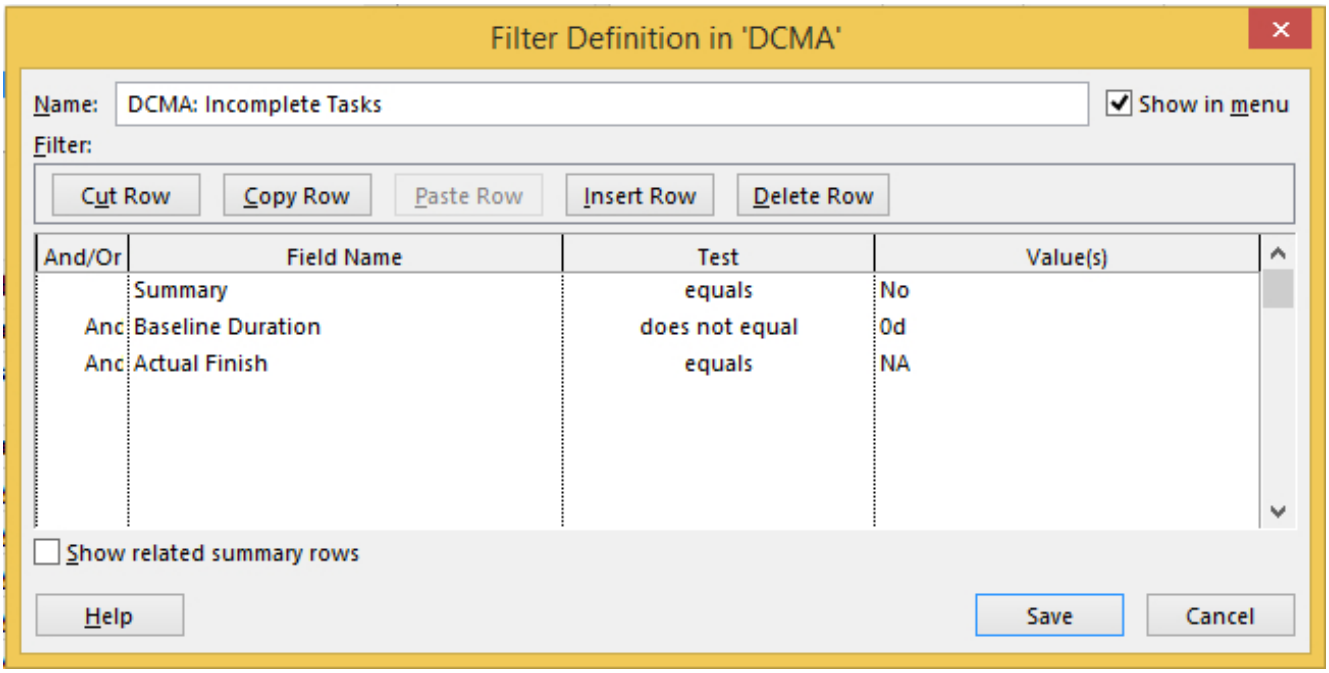
Incomplete Tasks Filter



	Task Mode	Total Slack	Task Name	Duration	Start	Finish	Predecessors
0		0 days	Sample Software Development Schedule	76.5 days	Tue 1/1/13	Wed 4/17/13	
1		30.5 days	Scope	46 days	Tue 1/1/13	Tue 3/5/13	
2		30.5 days	Determine project scope	45 days	Tue 1/1/13	Mon 3/4/13	4
3		0 days	Secure project sponsorship	1.5 days	Tue 1/1/13	Wed 1/2/13	
4		30.5 days	Define preliminary resources	1 day	Tue 3/5/13	Tue 3/5/13	2
5		13.5 days	Secure core resources	1 day	Tue 1/1/13	Tue 1/1/13	6
6		13.5 days	Scope complete	0 days	Tue 1/1/13	Tue 1/1/13	5 8FS+2 days
7		8.5 days	Analysis/Software Requirements	19.5 days	Wed 1/2/13	Tue 1/29/13	
8		13.5 days	Conduct needs analysis	10 days	Fri 1/4/13	Thu 1/17/13	6FS+2 days 9
9		13.5 days	Draft preliminary software specifications	3 days	Fri 1/4/13	Tue 1/8/13	8SS 1
10		13.5 days	Develop preliminary budget	2 days	Wed 1/2/13	Fri 1/4/13	9SF 1
11		108 hrs	Review software specifications/budget with team	4 hrs	Thu 1/3/13	Fri 1/4/13	10FF 1
12		13.5 days	Incorporate feedback on software specifications	1 day	Fri 1/4/13	Fri 1/4/13	11 1
13		8.5 days	Develop delivery timeline	1 day	Mon 1/14/13	Mon 1/14/13	12 1
14		68 hrs	Obtain approvals to proceed (concept, timeline, budget)	4 hrs	Tue 1/15/13	Tue 1/15/13	13 1
15		8.5 days	Secure required resources	10 days	Tue 1/15/13	Tue 1/29/13	14 1
16		8.5 days	Analysis complete	0 days	Tue 1/29/13	Tue 1/29/13	15 1
17		0 days	Design	43.5 days	Tue 1/1/13	Fri 3/1/13	
18		33 days	Review preliminary software specifications	2 days	Tue 1/29/13	Thu 1/31/13	16 9
19		33 days	Develop functional specifications	21 days	Thu 1/31/13	Fri 3/1/13	
20		33 days	Use Case 1	2 days	Thu 1/31/13	Mon 2/4/13	18 9
21		33 days	Use Case 2	3 days	Mon 2/4/13	Thu 2/7/13	20 9
22		33 days	Use Case 3	1 day	Thu 2/7/13	Fri 2/8/13	21 9

To build the Incomplete Tasks filter:

1. Select View – Filters – New Filter (Figure 1)
2. Rename the filter to Incomplete Tasks
3. Add the Summary field equals No
4. Add And Baseline Duration does not equal 0d
5. Add And Actual finish equals NA
6. Click Save



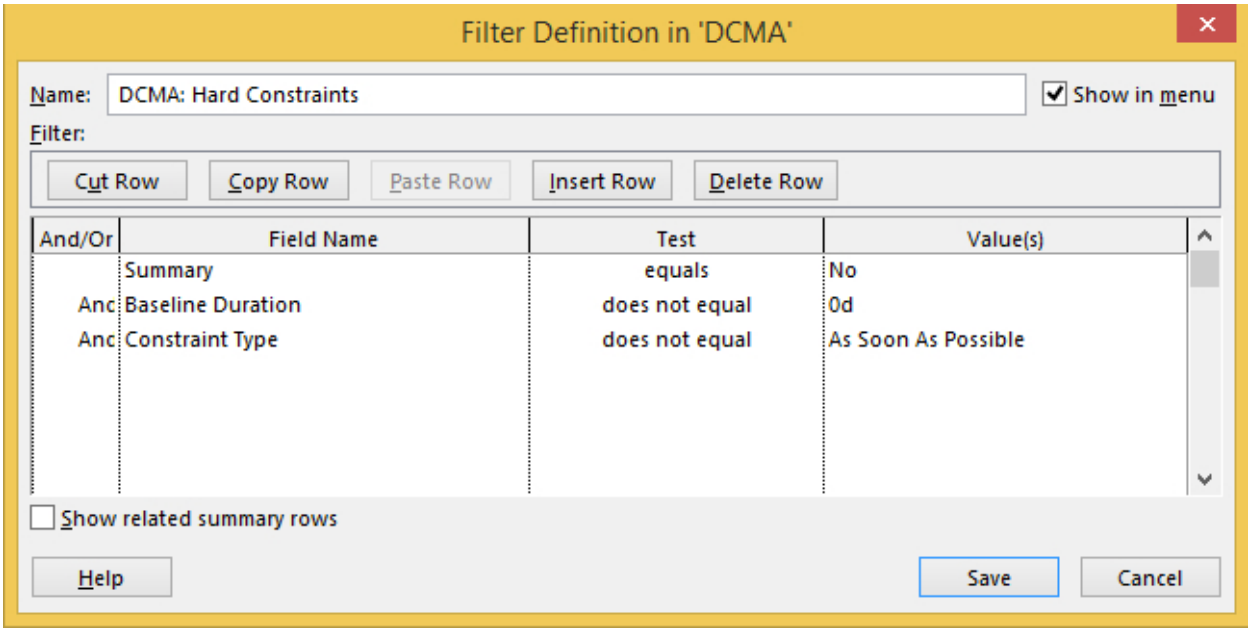
This filter will display all the incompleted tasks at the lowest level of the work breakdown structure. It strips out all the summary and milestone tasks so you can focus on the key tasks that represent the project schedule.

Hard Constraints Filter

A quality project schedule avoids hard constraints. The ideal schedule is dynamic and allows the logic to drive the forecast dates. Constraints such as Must Finish On, Must Start On, Start No Later Than or Finish No Later Than stop the logic from flowing through the schedule and often results in scheduling error messages. Therefore, all the constraints in the schedule should be set to Start As Soon As Possible.

To build the Hard Constraints filter:

1. Select View – Filters – New Filter
2. Rename the filter to Hard Constraints
3. Add the Summary field equals No
4. Add And Baseline Duration does not equal 0d
5. Add And Constraint Type does not equal As Soon As Possible
6. Click Save



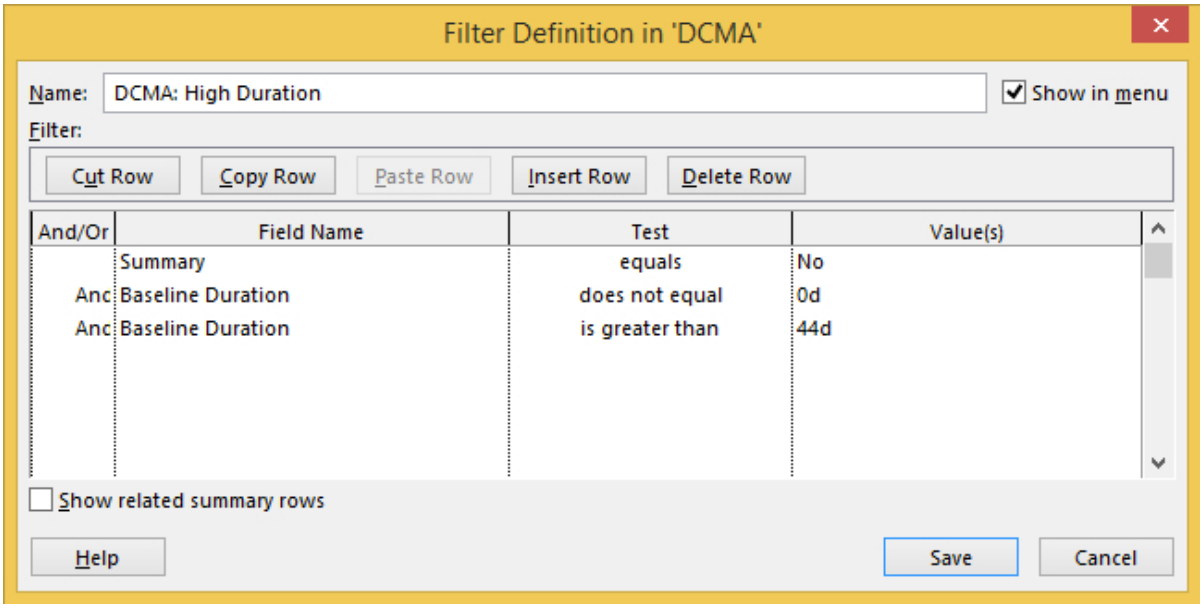
If the filter displays any tasks, you'll want to revisit the need for the constraint and consider using Predecessors to control the start of the task. It may be necessary to have hard constraints in the project schedule and if predecessors don't work for you, just remember to use the hard constraints sparingly.

High Duration Filter

The high duration filter identifies any task that has a baseline duration greater than 44 days (i.e. 2 months). As competent project managers, we know such tasks should be broken down into more discrete tasks. Trying to track progress against a 2 month task doesn't provide enough granularity to understand if the task will be delivered on time.

To build the High Duration filter:

1. Select View – Filters – New Filter
2. Rename the filter to High Duration
3. Add the Summary field equals No
4. Add And Baseline Duration does not equal 0d
5. Add And Baseline Duration is greater than 44d
6. Click Save



If the filter displays any tasks, you'll want to revisit the task duration and consider breaking the tasks down into smaller tasks with shorter durations.

These filters are just one approach to customizing Microsoft Project. You can also do more complex filter by creating custom fields and creating evaluation logic. By filtering on the custom field values, more complex filters can be developed. I'll show you how to develop more advanced filters in a future tutorial.